

Handbook of
**AGRICULTURE, ENVIRONMENT AND
BIOTECHNOLOGY**

— *Edited by* —

**A. Rakshit, H.B. Singh, A.K. Singh, P.C. Abhilash,
B.K. Sarma, V.K. Tripathi and H.P. Parewa**

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Different Initiatives under Agriculture Extension since Independence

**Riti Chatterjee, Suddhasuchi Das, Amit Baran Sharangi,
Sankar Kumar Acharya**

Agriculture is the part and parcel in the process of economic development in India. Besides providing food to nation, agriculture releases labour, provides saving, contributes to market of industrial goods and earns foreign exchange. Agricultural development is an integral part of overall economic development. In India, agriculture was the main source of national income and occupation at the time of Independence. Agriculture and allied activities contributed nearly 50 per cent to India's national income. Around 72 per cent of total working population was engaged in agriculture. These confirm that Indian economy was a backward and agricultural based economy at the time of Independence. After 61 year of Independence, the share of agriculture in total national income declined from 50 per cent in 1950 to 18 per cent in 2007- 08. But even today more than 60 per cent of workforce is engaged in agriculture. In spite of this, it is also an important feature of agriculture that is to be noted that growth of other sectors and overall economy depends on the performance of agriculture to a considerable extent. Because of these reasons agriculture continues to be the dominant sector in Indian Economy.

Since independence India has made much progress in agriculture. Indian agriculture, which grew at the rate of about 1 per cent per annum during the fifty years before Independence, has grown at the rate of about 2.6 per cent per annum in the post-Independence era. Expansion of area was the main source of growth in the period of fifties and sixties after that the contribution of increased land area under agricultural production has declined over time and increase in productivity became the main source of growth in agricultural production. Another important facet of progress in agriculture is its success in eradicating of its dependence on imported food grains. Indian agriculture has progressed not only in output and yield terms but the structural changes have also contributed. All these developments in Indian agriculture are contributed by a series of steps initiated by Indian Government. Land reforms, inauguration of Agricultural Price Commission with objective to ensure remunerative prices to producers, new agricultural strategy, investment in research and extension services, provision of credit facilities, and improving rural infrastructure are some of these steps.

In order to achieve the goal of self sufficiency in agriculture, new agricultural strategy has been initiated in 1966-67. The fundamental of this strategy is the application of science and technology for increasing yield per hectare. This strategy, known as New Agricultural Strategy or Green Revolution, is based on the extension of high yielding varieties responsive to heavy doses of fertilizers and the package of improved practices in selected areas with assured rainfall or irrigation facilities. The programmes included under the new strategy are: (1) the high yielding varieties programme, (2) multiple cropping programme, (3) integrated development of dry areas, (4) plant protection measures, (5) increased use of fertilizers, and (6) new irrigation concept.

Extension efforts since independence of India

The public agricultural extension system of India is one of the largest knowledge and information dissemination institutions in the world. The system played a critical role during the Green Revolution period; but in recent years, it has undergone a high level of scrutiny.

For the overall development of Indian agriculture, many institutional and infrastructural changes have been introduced since Independence. Broadly, agricultural policy followed during this period can be distinguished in four phases: first phase considered from 1947 to mid sixties, second phase considered period from mid- sixties to 1980, third phase included period from 1980 to 1991, and forth phase includes period from 1991/92 onwards.

The first phase of agricultural policy witnessed tremendous agrarian reforms, institutional changes, development of major irrigation project and strengthens of cooperative credit institution. The Community Development Programme, decentralised planning and the Intensive Area Development Programmes were also initiated for regenerating Indian agriculture that had stagnated during the British period. In order to encourage the farmers to adopt better technology, incentive price policy was adopted in 1964 and the Agricultural Price Commission was set up to advice the Government on the fixation of support prices of agricultural crops. Despite the institutional changes and development programmes introduced by the Government during this phase, India remained dependent upon foreign countries for food to feed the rising population. The second phase in Indian agriculture started in mid 1960s with adoption of new agricultural strategy. The new agricultural strategy relies on high-yielding varieties of crops, multiple cropping, the package approach, modern farm practices and spread of irrigation facilities. The biggest achievement of this strategy has been attainment of self-sufficiency in food grains. Agrarian reforms during this period took back seat while research, extension, input supply, credit, marketing, price support and spread of technology were the prime concern of policy makers. The next phase in Indian agriculture began in early 1980s. This period started witnessing process of diversification which resulted into fast growth in non-food grains output like milk, fishery, poultry, vegetables, fruits etc which accelerated growth in agricultural GDP during the 1980s. There has been a considerable increase in subsidies and support to agriculture sector during this period while public sector spending in agriculture for infrastructure development started showing decline in real term but investment by farmers kept on moving on a rising trend. The fourth phase of agricultural policy started after initiation of economic reform process in 1991. Economic reforms process involved deregulation, reduced government participation in economic activities, and liberalization. Although there is no any direct reforms for agriculture but the sector was affected indirectly by devaluation of exchange rate, liberalization of external trade and disprotection to industry. During this period opening up of domestic market due to new international trade accord and WTO was another change that affected agriculture. This raised new challenges among

policymakers. Because of this, a New Agricultural Policy was launched by Indian Government in July 2000. This aims to attain output growth rate of 4 per cent per annum in agriculture sector based on efficient use of resources. It seeks to achieve this objective in a sustainable manner and with equity. This was first time when government released a national agriculture policy. The policy document discusses what ought to be done in agriculture but the subsequent step, how and when policy goals and objective would be achieved is not discussed.

Extension Efforts in post-independence Era

In post independence era, government of India has organized extension activities in three distinct phases viz. extensive (i.e. Community Development and National Extension Service embraced all-round development), intensive (i.e. IADP, IAAP, HYVP) and Reformed Agricultural Extension (i.e. Training and Visit System). At present, extension work in the country is being carried out by five distinct organisation system.

1. National Extension Service (NES) is the main extension service, run by State Department of Agriculture, is also known as training and visit system of Agricultural Extension and co-ordinated by Ministry of Agriculture, Department of Agriculture and Co-operation at the centre.
2. The Indian Council of Agricultural Research (ICAR) and State Agricultural Universities (SAUs) Extension System also carry out limited extension work.
3. The public sector undertaking extension system which includes extension by undertaking in the co-operative sector and commodity specific extension undertaken by statutory commodity boards, with no coordination among themselves or linkages with other extension

Preliminary Governmental Efforts

Alber Mayar, an American rural development enthusiast introduced principles of agricultural extension in the first government-sponsored programme of rural development in Etawah launching Etawah Pilot Project (1948), U.P. The concept of multipurpose extension worker was originated from this project. Besides. Introduction of new agricultural technologies, organisation of cooperatives and programmes of health, sanitation, social education, etc. had been taken up.

Nilokheri experiment (1948) was another development project started under S.K. Dey for rehabilitation of the families displaced from West Pakistan due to partition. Vocational training-cum-production centre had been initiated It was an example of success in co-operation and community work.

These two projects were small experiments with lessons for planners and policy makers.

Community Development (CD)/National Extension Service (NES)

Community Development, a nationwide rural development programme launched during 1952. It was a programme of overall development promoting all sided development of the village community. Community development Blocks with about 100 villages in its jurisdiction were the centre of all development activities. A team of extension specialists belonging to rurally oriented disciplines under the leadership of Block development officers were appointed. Village level Extension workers were multipurpose extension agents at the grassroot level, though, it was from that the concept of Gurgaon

Project. A village guide had been posted to act as a channel through which the advice of the experts in various departments could be passed on to the villagers. The programme of introducing improved seed, implements, the methods of cultivation, etc. were started throughout the districts. In 1953, there was National Extension service implemented, which was an organisational structure for carrying out development work in the country.

Then, intensive approach to agricultural development was the call of 1960s in order to accelerate agricultural production rapidly. In 1960, Intensive Agricultural District Programme (IADP) was launched to provide inputs, credit, training and farm advice to formulate scientific farm plan. Thus, intensive extension was needed to transform traditional subsistence farming habits to one of improved agriculture. Later, new agricultural strategies were formulated like Intensive Agricultural Area Programme (IAAP) in 1964 and High Yielding Variety Programme (HYVP) in 1966 to help boosting agricultural production. Then State Agricultural Universities were started in every state to disseminate agricultural education. Farmers Training centres (FTC), National Demonstration (ND) scheme, farm and home broadcasting of All India Radio (AIR) and farmer's functional literacy programme were launched during 1965-66.

Target-Group Approach

Small Farmers Development Agency (SFDA), Marginal Farmers Development and Agricultural Labourers agency, Drought Prone Area Programme, Hill Area Development Programme, Tribal Area Development Programmes, etc. came up in 1970. These programmes were launched to meet the thrust of rural development programme in the country in neglected regions, in order to improve the conditions of small and marginal farmers, landless labourers, drought prone regions, hill areas. Then Integrated Rural Development Programme (IRDP) also implemented in 1976 for upliftment of the poor and oppressed group of people in rural areas and for generating income and employment. A series of schemes were launched to provide loans, subsidy and training on income generating skills and for creation of assets. The same approach was further strengthened in 1980s and 1990s. However, the extension functionaries and approach of extension remained unchanged. There was not much thinking on this aspect.

Training and Visit System (T & V)

To start with pilot projects were started in Rajasthan and Madhya Pradesh in 1974 the positive results inspired gradual expansion of the project throughout the country. The training and visit system called for major overhaul of the existing agricultural extension system by initiating reorganisation at all levels. The new system called for single line of command from the state headquarters to the grass root level. The extension personnel were made responsible only for professional agricultural extension works. Systematic visits are conducted soon after training to disseminate messages. At the grassroots level, contact farmers are selected (almost 10 per cent out of total farm families), they serve as key communicators. The village extension workers (grassroot level functionary) contact those selected farmers on their visits. There is close and constant interaction with the research system. The State Agricultural Universities and their outreach stations do not only conduct monthly workshops for subject matter specialists but also take interest in research problems based on needs of the farmers. There is provision for joint visits of university scientists, and extension officers to the farmers' fields from time to time.

A Village Extension Worker (VEW) form a group of 500 farm families in intensive areas like major irrigation systems or areas with large distance or having other special considerations and for a group of 800 farm families in the unirrigated areas. One Agricultural Extension Officer (AEO) is provided for VBW at the block level.

Sub-divisional Level- A sub-division covering about 3-5 blocks, is the basic unit for the operation of this system. Since the sub-division is the responsible unit for training for the extension workers, it is supported by a team of Subject Matter Specialists (SMS), especially in the fields of Agronomy and Plant Protection along with the Training Officer. This unit is again headed by the Sub-divisional Extension Officer.

District Level- A district headed by a District Extension Officer and assisted by a team of Subject Matter Specialists covering 3-5 subdivisions is normally the unit for implementation of programmes at this level.

Zonal Level- In case of too many districts to be supervised directly by the headquarters, one additional level has been provided, the zonal level. The Zonal Extension Officer (Z. E.O.) is responsible for all extension activities in the districts under him. In other matters, the links between the department and the districts will be through the ZEO.

State Level- The extension service is placed under a whole time Additional Director of Agriculture (Extension) with suitable technical assistance under the overall charge of the Director of Agriculture. The A.D.A. (Extension) is fully responsible for the implementation of Extension activities, in the state he is assisted by a team of most senior SMSS.

The key features of Training and Visit system are as follows

Professionalism, single line of command, time bound work, regular and continuous training and linkage with research.

Achievement and Weakness of Training and Visit system

Though this system was quite successful in increasing cultivated area and HYV, disseminated new technologies among the farming community, enhanced farm employment, but limited use of mass media made hindrance in reaching small and marginal farmers.

First Line Extension Projects of the Indian Council of Agricultural Research

The Indian Council of Agricultural Research, an apex organisation for co-ordinating agricultural research and education in the country, has a limited role in extension. It has launched a few programmes popularly known as transfer-of-technology projects or First-Line Extension Projects. The major goal behind launching these programmes is to demonstrate latest agricultural technologies in the farmers' fields and gain first-hand feedback by direct interaction between scientists and farmers. It was believed that experiences gained in such sound extension programmes would help in designing model extension programmes to be implemented by general extension system. Brief description of each programme is given below.

Lab to Land Project

Lab-to-Land project was launched as part of the ICAR Golden Jubilee programme in 1979. It aimed to bring scientists and farmers in close contact and demonstrate potentials of low cost technologies relevant for small marginal farmers as well as landless labourers. The emphasis was made on developing farm plan, keeping in view resources available, technologies appropriate for diversification of labour use and creating supplementary sources of income. The programme was implemented through State Agriculture Universities, ICAR Institutes, State Departments of Agriculture, Voluntary Organizations. The activities under Lab-to-Land project included creation of farm plan, and training, and demonstration. The whole programme was acted as feedback mechanism for the scientists and extension functionaries.

Krishi Vigyan Kendra (Farm Science Centre)

Krishi Vigyan Kendra (KVK) was launched in 1974 as vocational training institution for farmers and field-level extension functionaries. It aimed to impart need-based and skill-oriented training to enhance agricultural production. Training courses of each Kendra should be tailored according to felt-needs and existing resources of particular area. Three fundamental principles or mandates of KVK are as follows:

- (i) Agricultural production is the prime-goal,
- (ii) Work experience is the main method of facilitating training, and
- (iii) Priority of training should be on the weaker section of the society.

The KVKs are now engaged with four major functions, as given below:

- (i) To organise long-term vocational training for rural youths for generating self-employment,
- (ii) To organise training of farmers and extension functionaries,
- (iii) To organise front-line demonstrations, and
- (iv) To collaborate with subject matter specialists (SMS) of Agricultural Universities in on-farm testing, refining and documenting technologies.

National Demonstration Project (ND)

National Demonstration Project (1964) had mainly taken care of multiple cropping demonstration, Rainfed cultivation demonstration, Special problem soil demonstration, Entire farming system demonstration.

Operational Research Project (ORP)

Operational Research Projects (ORP) were initiated in 1975 by ICAR with the aim of demonstrating the technologies to the farmers and the extension workers on a watershed basis to evoke community action. The ORP had the following specific objectives:

- (i) To test, adopt and demonstrate new agricultural technology on farmers' fields in the whole village or cluster of contiguous villages/water area,
- (ii) To determine profitability of the new technologies and their place of spread among farmers.
- (iii) To demonstrate group action as a method of popularising the most of the technologies at faster rate.

Institution Village Linkage Programme (IVLP)

Institution Village Linkage Programme (IVLP) was started on a pilot basis in 1995 as a programme of the ICAR emphasizing on technology integration. The technology integration in contrast to technology transfer, refers to the process or a set of activities that must happen if a new scientific agricultural information is put to use in any farm production system. In view of three districts types of agriculture, viz. commercial green revolution and complex, diverse and risk prone (CDR), there is need for technological intervention through technology assessment and refinement with the emphasis on stability and sustainability for small production system, maintaining productivity and profitability with environmental concerns for well-defined farm production systems and appropriate for value addition in case of commercial farm production system. This was implemented in 42 centre of selected ICAR institutes and SAUs. Thus, IVLP is a programme generating appropriate technologies by refining and assessing innovation generated by scientists knowledgeable about their respective disciplines. However, various factors operating in the farm production system of small farmers hitherto not considered by scientists, affect adoption process IVLP being a 'bottom up' approach, therefore, has following specific objectives:

- (i) To introduce technological intervention with emphasis on stability and sustainability along with productivity of small production systems.
- (ii) To introduce and integrate the appropriate technologies to sustain technological interventions and their integration to maintain productivity and profitability taking environmental issues into considerations in a comparatively well-defined production system,
- (iii) To introduce and integrate the appropriate technologies to increase the agricultural productivity with marketable surplus in commercial farm production systems.
- (iv) To facilitate adoption of appropriate post-harvest technologies for conservation and on-farm value addition of agricultural products, by-products and wastes for greater economic dividend and national priorities
- (v) To facilitate adoption of appropriate drudgery, increased efficiency and higher income of farm women.
- (vi) To monitor socio-economic impact of technologies for removal of the technological intervention(%)for different production systems.
- (vii)To identify extrapolation domains for new technology/technology modules based on environmental characterization at meso and mega levels.

National Agricultural Technology Project (NATP)

In 1999, Indian Council of Agricultural Research (ICAR) launched National Agricultural Technology Project (NATP). Under this project. Innovations in technology Dissemination (ITD) component has been planned, which aims at developing transfer of technology (TOT) system that is demand driven, well integrated with research, and financially sustainable and accountable to stakeholders of agricultural development.

Objectives of NATP

1. To increase the quality and type of technologies disseminated,
2. To enable research and extension systems to become demand driven and responsible for solving problems of farmers,
3. To strengthen research-extension-farmers (R-E-F) linkages,
4. To increase financial sustainability of the system,
5. To share ownership of Agricultural Technology System (ATS) among key stakeholders ie, farmers especially poor, women and disadvantaged, public sector research and extension agencies like ICAR, SAUS, DOA etc.

National Agricultural Innovation Project (NAIP)

The World Bank assisted agricultural research programme called the National Agricultural Innovation Project (NAIP) was launched in India in July, 2006, for a period of six years. The project focuses on innovations in agricultural technology. The project would facilitate an accelerated and sustainable transformation of the Indian agriculture so that it can support poverty alleviation and income generation. This would be achieved through collaborative development and application of agricultural innovations by public organisations in partnership with farmers' groups, the private sector and their stakeholders.

Objectives of NAIP

The important objectives of NAIP are:

1. To build the critical capacity of the Indian Council of Agricultural Research (ICAR) as a catalyzing agent for management of change of the Indian National Agricultural Research System (NARS),
2. To promote research in the production-to-consumption system mode in the priority areas/themes to enhance agricultural productivity and profitability and nutrition, income and employment of the rural stakeholders.
3. To improve livelihood security of the rural people living in selected disadvantaged regions through technology-led innovative systems which encompass the wide process of social and economic change among all the stakeholders,
4. To build capacity to undertake basic and strategic research in strategic areas agricultural sciences, to meet the technology development challenges in the immediate and predictable future.

Key components of NAIP

The key components of NAIP are as follows:

Component-1: Institutional development by strengthening of ICAR role,

Component-2: Establishment of consortia for research on production to consumption system,

Component-3: Establishment of consortium for research on sustainable rural livelihood security in disadvantaged areas,

Component-4: Establishment of consortia to strengthen basic and strategic research in areas of agricultural science.

Selection of disadvantaged regions

One hundred and fifty disadvantaged districts have been identified by the Planning Commission covering 27 States and 13 agro-climatic zones under NAIP.

Three variables were considered for computing the index of backwardness, they are as follows:

- (i) Agricultural productivity per worker,
- (ii) Agricultural wage rate, and
- (iii) SC/ST population

One hundred and fifty districts have been grouped into 13 clusters. Each cluster consisting of contiguous districts of similar socio-economic and agro-ecology will be preferred for consortia formation.

National Food Security Mission (NFSM)

The World Bank (1986) defined food security as the access by all people at all times to enough food for an activity, healthy life. Its essential elements are availability of food and the ability to acquire it.

In view of the stagnating food grain production and an increasing consumption need of the growing population, Government of India launched this centrally sponsored scheme, National Food Security Mission (NFSM) in August 2007.

The major objective of the scheme is to increase production and productivity of wheat, rice and pulses on a sustainable basis so as to ensure food security of the country. The Approach is to bridge the yield gap in respect of these crops through dissemination of improved technologies and farm management practices.

NFSM will have three components

- ☐ National Food Security Mission-Rice (NFSM -Rice)
- ☐ National Food Security Mission - Wheat (NFSM - Wheat)
- ☐ National Food Security Mission - Pulses (NFSM - Pulses)

Various activities of NFSM were undertaken related to demonstration of improved production technologies, distribution of quality HYV and hybrid seeds, popularization of newly released varieties, support for micro-nutrients, gypsum, zero-tillage, rotavators, weeders and other farm implements, IPM, INM, extension, water lifting and moisture saving devices, training and mass media campaign including awards for best performing districts.

Rashtriya Krishi Vikas Yojana (RKVY)

A new State Plan Scheme of Additional Central Assistance for agriculture and allied sectors, namely, *Rashtriya Krishi Vikas Yojana* (RKVY) was launched during 2007-08, with an envisaged outlay of

25,000 crore for the 11 Five Year Plan. The broad objective of the scheme is to provide incentive to the States to increase public investment to achieve 4 per cent growth rate in agriculture and allied sectors in the 11 Five Year Plan. Under the scheme, the following broad activities have been identified for focus attention Integrated development of food crops, including coarse cereals, minor millets and pulses Agriculture mechanisation; Soil health and productivity: Development of rainfed farming systems; Integrated pest management; Market infrastructure, Horticulture Animal husbandry and dairying; Fisheries: Timely completion of projects; Support to institutions that promote agriculture horticulture etc: Organic and bio-fertilizers, and innovative schemes. These activities are only indicative in nature.

Swarnajayanti Gram Swarozgar Yojana (SGSY)

The *Swarnajayanti Gram Swarozgar Yojana* (SGSY) was launched in India on 15 April 1999 based on review of prof. S. R. Hashim Committee of the neural development and poverty alleviation programmes in 1997. This committee, among others, suggested for making efforts towards from Individual beneficiaries approach to soup approach in implementation of self generation programme and preference was given to group activities and cluster approach. SGSY is actually the conglomeration of the previously lunched programmes such as Integrated Rural Development Programme (IRDP) and Training of Rural Youth for Self-employment (TRYSEM), Development of Women and children in Rural Areas (DWCRA), Supply of Improved Toolkits to Rural Artisans (SITRA). *Ganga Kalyan Yojana* (GKY) and Million Well Scheme (MWS). This holistic SGSY programme encompassing all aspects of self-employment such as organisation of the poor into self-help groups, training, credit, infrastructure and marketing was evolved as a means of poverty alleviation in rural areas.

Objectives

The objective of the programme is to bring the assisted poor families (*Swarozgar*) above the poverty line by providing them income generating assets through a mix of bank credit and government subsidy

Features-

- (i) Establishing a large number of micro-enterprises in rural areas building upon the potential of the rural poor.
- (ii) Assisted families may be individuals or groups but the emphasis is on the self-help groups.
- (iii) Cluster approach for establishing micro-enterprises, identification of 4 to 5 activities in each block, based on resources, occupational skills of the people and availability of market.
- (iv) Project approach for each key activity.
- (v) SGSY is a credit-cum-subsidy programme.
- (vi) Emphasises skill development through well designed training programmes.
- (vii) Promotion of marketing of the goods produced by *swarozgaris*.
- (viii) Special focus on vulnerable groups - SC/ST at least 50 per cent, women 40 per cent and disabled 3 per cent.
- (ix) Funds for SGSY is shared between Central and State Governments in the proportion of 75:25

Pradhan Mantri Gramin Sadak Yojana (PMGSY)

Pradhan Mantri Gramin Sadak Yojana (PMGSY), a centrally sponsored scheme, launched on 15th August 2000, is connecting the India till date.

Objectives

1. To provide easy accessibility to the rural community,
2. To conduct rural development,
3. To improve socio-economic condition of the rural population,
4. To bridge the gap between urban and rural areas.
5. To connect each and every village with a population exceeding 1000 by all-weather roads.
6. To maintain rural roads constructed under this scheme by the local *panchayats* with the help of State government fund.

Sampoorna Gramin Rozgar Yojana (SGRY):

Sampoorna Gramin Rozgar Yojana (SGRY), a centrally sponsored scheme into which the two schemes, namely, Employment Assurance Scheme (EAS) and *Jawahar Gram Samridhi Yojana* (JGSY) administered by the Panchayats and Rural Development Department have been merged effect from 2001-02.

Objectives

1. To provide additional wage employment and food security in the rural area.
2. To create of durable community, social and economic assets, and
3. To develop rural infrastructures.

Mahatma Gandhi National Rural Employment Guarantee Act (MGNREGA):

The Parliament passed the National Rural Employment Guarantee Act (NREGA) 2005 that guarantees 100 days of wage employment in a year to every rural household whose adult members are willing to do skilled manual work. The Act has become operative in the notified districts from 2nd February 2006. In the first phase, 200 districts were identified for implementation of NREGA. With effect from 1st April, 2008, NREGA has been extended to all districts in the country. NREGA is renamed as Mahatma Gandhi National Rural Employment Guarantee Act (MGNREGA) on 2nd October, 2009.

Objectives

The main objective of the MGNREG Act is to provide for the enhancement of livelihood security of the households in rural areas by providing at least one hundred days of guaranteed wage employment in every financial year to every household whose adult members are willing to do unskilled manual work (The Gazette of India Extraordinary 2006: 1).

There are other objectives also, such as:

1. Reduction of number of distressed migration from the rural to urban areas and from one part of rural to another part of rural areas,
2. Creation of durable assets in rural areas,
3. Empowering rural women through opportunity to earn income independently and to participate in social groups,
4. Overall development of the rural economy,
5. Promotion of inclusive growth and development.

Pradhan Mantri Fasal Bima Yojana (PMFBY)

In January, 2016, the government of India *launched Pradhan Mantri Fasal Bima Yojana* (PMFBY), a crop insurance scheme, which covered risks of weather, pests, diseases, prevented sowing and damages on standing and harvested crops. Although government continued to subsidize premiums paid by farmers, empanelment of private insurers for quoting competitive actuarial rates invited private sector participation. Modern technology accelerated the pace and intensity of collection of yield data so that the notified 'Area' can become truly homogeneous in future. Crop insurance can be a critical protection against climate change but a weather calamity can strike a vast area and large numbers. The nation's satellite and meteorological capabilities are strengthened to generate early and reliable alerts. A disaster response force was instituted with passage of a Disaster Management Act in 2005 whereby Central forces are deployed to support the states as and when needed. Drought is treated as a disaster calling for special subsidies on irrigation when state notifies drought.

Role of State Agricultural Universities (SAUs) in Extension

State Agricultural Universities (SAU) were set up in India on the pattern of Land Grant Universities of U.S.A. with integrated teaching, research and extension functions. Each SAU was expected to perform some extension roles in the area under its jurisdiction. However, extension role of SAU was supposed to be complementary to the major extension efforts made by State Department of Agriculture. The specific extension roles of SAUs are as given below.

- (i) To prepare personnel to perform various extension roles and provide dynamic and analytic leadership to extension programmes through a sound, systematic and integrated undergraduate and post-graduate teaching in extension education.
- (ii) To collect, process and disseminate latest research findings to extension personnel and extension clientele through appropriate methods and media
- (iii) To identify field problems to which scientific solutions are Available and communicate these to the concerned subject e departments
- (iv) To produce information materials and teaching aids for extention personal, trainers and farmers.
- (v) To plan, organise and conduct production and problem oriented training of various types and duration for extension personnel, farmers and others concerned with development programmes
- (vi) To provide authentic and systematic guidance through an uptodate and specialised Farm and Home Advisory Service to farmers, farm women, farm youth, extension personnel, bankers,

input dealers manufacturers, voluntary farm organisations and others concerned with agricultural development.

(vii) To undertake researches on various facets of extension education with a view to enriching programme formulation and operation, and gaining insight into the process of dynamics of planned change.

(vii) To undertake adaptive and operational research projects with a view to testing applicability of findings under different soil, agro-climatic and socio-economic conditions in the different parts of the state.

In order to assist and perform these roles each SAU must have Directorate of Extension with following major units.

Farm Advisory Services

Extension workers and farmers need advice on technical matters such advice should be readily available with the help of concerned subject matter specialists with the help of SAUs.

Training

Training of farmers, farm women, youth and extension personnel has become one of the major responsibilities of SAUs and appropriate infrastructure have been created for the purpose.

Field Extension

Each SAU undertakes field extension programmes in the areas under its jurisdiction through subject matter specialists of Directorate of Extension or regional stations. Besides the transfer of technology programmes of ICAR viz. National Demonstration, KVK, LLP, ORP and Technology.

Assessment and Refinement through Institutional Village Linkage Programme (IVLP) are also implemented. Some SAUs have well planned field extension programmes of their own.

Extension Role of Voluntary Organisations

Voluntarism is derived from latin word '*Voluntas*' which means will. The free will assumes various forms of impulses, passions and desire to act. The term 'volunteer' is normally used to mention someone who offers unpaid service for a noble cause. There are many sources of motivation behind voluntarism like religion, love of mankind, or mutual aid. Voluntary organisation refers to non-profit organisations set up by a group of persons on voluntary basis to help people of a locality in self-reliant development in order to satisfy their needs. Voluntary organisations must be representatives of the needs and aspirations of the people and should not expect to get in return for the services rendered. It must also be free from external control. People's involvement in all the activities is essential. Ideally the emphasis should be on raising consciousness of the people and strengthening their capacities to solve their own problems.

New Dawn in Indian Extension

Extension work is undergoing a phase of transition due to various factors such as technological innovations, new age media, changes in organisations and perception about development.

Thus, many new trends are emerging such as inclusion of new subject matter area, in the area of extension education, privatisation, media use, participatory methods of extension work, farmers network, that will be discussed below:

New Subject Matter Areas

Extension though started with improvement in agriculture and home economics in early years, it soon embraced other subject matter e-rural living viz. fisheries, forestry, livestock, veterinary, dairy concerning etc. However, with changing demography in the developed nations bulk of population in Europe and North America is urban. Extension in many of these countries is now focussing on new subject matter areas of relevance to the majority viz. environment, energy, social-ecology, occupational health hazards of both farmers and farm women, consumer education, natural resources management, etc. With success of extension in public education, many democratically elected governments find it natural to propagate public awareness about many new areas like family welfare, teenage pregnancy, drug prevention, care of the elderly through time tested methods of extension education. The World Bank supported extension is broadening its narrow focus from crops to livestock, land use, water-management, conservation of natural resource, climate-smart agriculture, and precision agriculture in order to provide much needed technological support to farmers

Privatisation of Extension

Extension is increasingly being seen as a profitable resource having commercial value. Thus, there is growing tendency in developed nations towards making agricultural extension into a profitable enterprise. It is believed that clients should pay partially to fully for self-financing of extension work. With resource crunch and financial crisis being encountered by many national governments, fund generation is becoming highly needful day by day. Many large scale farmers and agribusiness organisations can willingly pay for the advice and assistance received from extension organisations. In Britain, the main goal of its extension service is to be leading consultancy to farmers and agribusiness to provide quality service: Of late, Government of Rajasthan State of India has decided to involve private sector voluntary agency to undertake extension at the grassroot level and gradually minimise government's role in extension work. This new trend is giving rise to misperception that public-funded extension service is biased towards those who can afford the high price. With increasing emphasis on profit, the attention would now turn to physically well endowed regions, cash crops and big farmers. Further, the public organisation would be weakened in comparison to private organisation.

Media Use

Agricultural extension, for long, has depended on traditional modes of communication, mostly face-to-face contacts with individuals and groups despite the fact that number of extension workers is woefully inadequate in many parts of the world. Many experts feel that traditional extension is labour intensive and not well-equipped to meet diverse information needs to all categories of farmers. Communication media, offer a lot of advantages for reaching messages to masses more precisely and quickly. Above all, range of communication media has expanded far beyond imagination. There are exciting new features available in the new communication technologies combining both capacity of broadcast

and miniaturization of equipments in order to facilitate grassroots level production and distribution. Availability of computer and telecommunication facilities are opening new streets of communication

Participatory Approaches to Extension

Extension work being carried out by bureaucratic government organisations under strict mandates are seen as anti-thesis of human emphasis on people-oriented extension organisations at the grassroots with emphasis on effective use of community's resources. The knowledge and experiences of people should be the starting point to solve community's problems. Farmers are being viewed as active partners in development of innovations and their dissemination in the community. Farming Systems Research and Extension (FSRE), Participatory Technology Development (PTD) and Farmer First extension are new waves of extension calling for active roles of farmers and building structures for farmer to farmer extension. Extension in the changed scenario is more farmer-propelled and interactive.

Establishment of Networks among Indian farmers

Establishment of links among farmers with similar concerns and mission together in search of solutions of the problems is becoming the call of decade. As these networks can provide such forum to people with overlapping vision, and need to maintain contact. This is more relevant for wt reliance of farmers, especially to those not having ready access advice. They can use their network to gather time-to-time needed information. Before starting, it requires some baseline survey to determine the need of potential member and extent of commitment to work together for the common cause. Depending upon the objectives, formal or semiformal structure need to be formulated. A steering committee or administrative body should be formed to pull fund, distribute responsibilities and carry out functions. In India, a network of small and marginal farmers interested in LEISA (Low External Input Sustainable Agriculture) was formed in Tamil Nadu and Pondicherry in 1990. The organisation for network was initiated through a regional workshop of farmers with common interest.

Lessons learnt from experiences of farmers network in developing nations indicate that, initially an informal organisational structure with focus on actions are perceived as important by members. Then autonomy of members and their representation should be considered. Type of activities chosen for the network should be within limits of time and energy so as to allow for frequent meetings and sharing.

Extension Education in General Universities

Extension has now become part and parcel of agricultural higher education in India. General universities had so far been concentrating on teaching and research alone, confining their activities within four walls. With initiatives of pioneering thinkers and policy makers, extension has now become third and important dimension for general education. There are 95 universities having departments or centres of adult, continuing and extension education (ACEE) all over the country. Though all the departments, centres have not taken up activities equally intensively, some beginning has already been made. There is a general fear and hitch among the educated elites about content and manner of contact with the masses. However, there is no dearth of subject matter areas to be covered following 'cafeteria approach' under extension programmes. Extension seeks to disseminate knowledge for the benefit of community. Thus, concepts of home science, environment, commerce, botany, economics and management, law,

sociology, social work, etc. can be directly used to address to the information needs of the people. However, there is need to orient the teachers about their extension roles and nature of activities to be undertaken by the ACEE Units.

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